**Definition 2.6**

Let $r$ and $\theta$ be polar coordinates of the point $P(x, y)$ that corresponds to a non-zero complex number $z = x + iy$. The polar form or trigonometric form of a complex number $P$ is

$$z = r(\cos\theta + i\sin\theta).$$

For convenience, we can write polar form as

$$z = x + iy = r\left(\cos\theta + i\sin\theta\right) = r\,cis\,\theta.$$

The value $r$ represents the absolute value or modulus of the complex number $z$. The angle $\theta$ is called the argument or amplitude of the complex number $z$ denoted by $\theta = \arg(z)$.

(i) If $z = 0$, the argument $\theta$ is undefined; and so it is understood that $z \neq 0$ whenever polar coordinates are used.

(ii) If the complex number $z = x + iy$ has polar coordinates $(r, \theta)$, its conjugate $\bar{z} = x - iy$ has polar coordinates $(r, -\theta)$.

Squaring and adding (1) and (2), and taking square root, the value of $r$ is given by $r = |z| = \sqrt{x^2 + y^2}$.

Dividing (2) by (1), $\dfrac{r\sin\theta}{r\cos\theta} = \dfrac{y}{x} \Rightarrow \tan\theta = \dfrac{y}{x}$.

Case (i) The real number $\theta$ represents the angle, measured in radians, that $z$ makes with the positive real axis when $z$ is interpreted as a radius vector. The angle $\theta$ has an infinitely many possible values, including negative ones that differ by integral multiples of $2\pi$. Those values can be determined from the equation $\tan\theta = \dfrac{y}{x}$ where the quadrant containing the point corresponding to $z$ must be specified. Each value of $\theta$ is called an *argument* of $z$, and the set of all such values is obtained by adding multiple of $2\pi$ to $\theta$, and it is denoted by arg $z$.

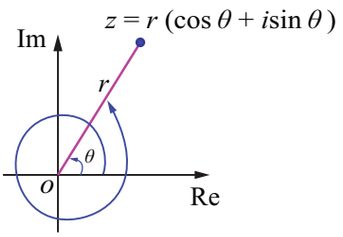



Fig. 2.29

Case (ii) There is a unique value of $\theta$ which satisfies the condition $-\pi < \theta \leq \pi$.

This value is called a principal value of $\theta$ or principal argument of $z$ and is denoted by Arg $z$.

Note that,

$$-\pi < \text{Arg}(z) \leq \pi \quad or \quad -\pi < \theta \leq \pi$$

Principal Argument of a complex number

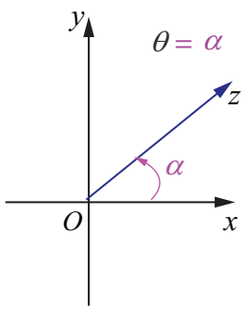
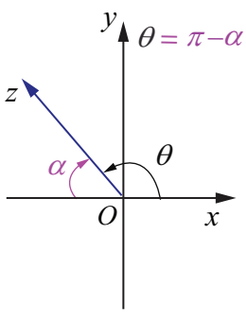
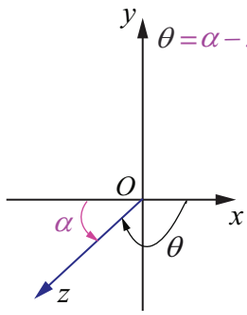
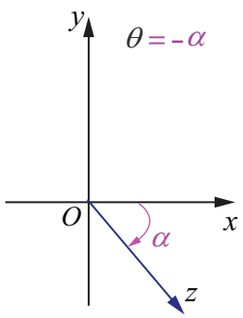
| I-Quadrant | II-Quadrant | III-Quadrant | IV-Quadrant |
|---|---|---|---|
|  |  |  |  |
| $\theta = \alpha$ | $\theta = \pi - \alpha$ | $\theta = \alpha - \pi$ | $\theta = -\alpha$ |



Fig. 2.30



Fig. 2.31



Fig. 2.32



Fig. 2.33



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